

Jack Gerdson

jgerdsen@andrew.cmu.edu | 908-343-9107 | Pittsburgh, PA | U.S. Citizen, no sponsorship required
linkedin.com/in/jackgerdsen | jacksongerdson.com | github.com/jackgerdsen

SUMMARY

Robotics software engineer working in autonomy, perception, and control. Carnegie Mellon robotics undergraduate, Software Controls Lead on 80-person NASA Lunabotics team, and teaching assistant for 16-385 Computer Vision, with ROS 2, simulation, and teleoperation stacks shipped across three CMU research labs. Seeking summer 2027 robotics software engineering internship.

EDUCATION

Carnegie Mellon University, School of Computer Science

Pittsburgh, PA | Expected May 2028

Bachelor of Science in Robotics, Minor in Machine Learning

GPA: 3.5/4.0

Relevant Coursework: Computer Vision, Robot Planning Techniques, Intro to Machine Learning, Robot Kinematics, Intro to Feedback & Control Systems, Intelligent Robot Systems, Computer Systems, Matrices & Linear Transformations, Robot Mathematics, Mathematical Foundations of Computer Science, Great Ideas of Theoretical Computer Science

SKILLS

Languages: C++, Python, C, Swift, Kotlin

Robotics: ROS2, Gazebo, MuJoCo, teleoperation, manipulation, motion planning, kinematics, feedback control

Perception & ML: SLAM (Simultaneous Localization and Mapping), computer vision, LiDAR, point clouds, sensor fusion, probabilistic state estimation, Kalman filtering, PyTorch, machine learning, dataset curation

Tools: Linux, Git/GitHub, Docker, NumPy, Nav2, RViz, Zenoh, Jetpack Compose, OpenCV, MoveIt, Isaac Sim

EXPERIENCE

Kantor Lab, CMU | Research Assistant

Pittsburgh, PA | Jun 2026 – Present

- Build gripper-mounted capture rigs used by lab members to collect vineyard field imagery in California.
- Curate imagery into labeled Python training sets, owning data pipeline behind models locating grape clusters.
- Define label schema and training targets for 18-month, \$1M project, setting the standard for data collection.

Carnegie Mellon University | Teaching Assistant, 16-385 Computer Vision

Pittsburgh, PA | Aug 2026 – Present

- Hold office hours for 80 students as one of four teaching assistants on image processing and geometric vision.
- Redesign assignments around written derivations, keeping 16-385 rigorous as AI use in coursework grows.

Momentum Robotics Lab, CMU | Research Assistant

Pittsburgh, PA | Jun 2025 – May 2026

- Built modular iOS teleoperation interface in Swift driving various 7-DoF manipulator from one controller, replacing separate apps per arm model.
- Packaged teleoperation stack with Zenoh and Docker on Linux so operators ran it off-site without rebuilding from source.
- Validated real-time control and fail-safes in MuJoCo, gating every hardware run on passing safety tests.
- Co-authored teleoperation section of 6-author paper submitted to RSS 2027.

CREATE Lab, CMU | Research Assistant

Pittsburgh, PA | Feb 2025 – May 2025

- Developed Android UI in Jetpack Compose communicating over Bluetooth with devices for MindfulNest project.
- Refactored Kotlin apps behind interactive toys for children, halving input latency and raising classroom engagement.
- Deployed Pre-K learning centers across Allegheny County, gathering structured teacher feedback at each site.

LEADERSHIP & ACTIVITIES

CMU Moon Miners (NASA Lunabotics) | Software Controls Lead

Pittsburgh, PA | Dec 2025 – Present

- Lead software for 80-person team building autonomous lunar excavation rover for NASA Lunabotics.
- Direct 20 software members across sub-teams, owning GitHub workflow, code review, and release deadlines.
- Delivered first autonomous scoring run by any rookie team in Lunabotics history, earning Best Rookie Team 2026.
- Wrote C++ and Python ROS 2 autonomy foundation on colcon build so every sub-team runs identical nodes in simulation and on hardware.
- Integrate LiDAR point-cloud SLAM for obstacle detection, localization, and autonomous navigation.

CMU RoboClub | Member, Simulation and Controls

Pittsburgh, PA | Sep 2024 – Present

- Created simulation adapters redirecting ROS 2 components into Gazebo, letting members test control code with no hardware.
- Modeled dynamic quadrupeds in Gazebo to prototype gait and balance controllers before anything was fabricated.

Fringe Buggy | Pusher

Pittsburgh, PA | Aug 2025 – Present

- Race unpowered aerodynamic buggies in CMU intramural competition, one of oldest traditions on campus.
- Organize on-campus events bringing broader university community out to races each semester.